19/10/2026

Week - 5

Encapsulation :- In python it means wrapping data (variable) and methods (functions) together inside a class and controlling access to that data.

→ oop (Object - Oriented programming)

Class :- A class is a blueprint or template used to create objects. It defines the attributes (data) and methods (functions) that objects can have.

eg :- A student class can define student name, date of birth, programmes, marks ; and methods to display information or check grades.

Object :- An object is an instance of a class. We create an object by calling the class name with parenthesis.

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22/10/202

Self Notes

week 5 - classes and objects

① what is a class ?

⟶ A class is a blueprint used to create objects

PYTHON CODE :

```
class student :
    name = "Aryan"
```

② what is an object ?

⟶ An object is an instance of a class.

PYTHON CODE :

```
class student :
    name = "Aryan"
s1 = student()
print (s1.name)
```

OUTPUT :

Aryan

③ what is init ( ) ?

→ - init - ( ) is a special method that runs automatically when an object is created. It initializes the object's attributes.

PYTHON CODE :

```
class student :
    def - init - ( self, name ) :
        self . name = name
S1 = Student ("Aryan")
print ( S1 . name )
```

Output :

Aryan.

In short - difference (Imagine you are making a student record)

① class - Blueprint ⇒ It difference how to create something
Example : student

② object - Actual thing ⇒ An object is create using a class
Example : S1 and S_2

③ - init_ ( ) - setup ⇒ It initializes each new objects data.
Example : assigning a student's name.

④ what is self ?

→ self refers to the current object

PYTHON CODE :

```
class students :
def __init__ (self, name):
    self.name = name
def show (self):
    print (self.name)
S1 = students ("Aryan")
S2 = students ("Rahul")
    S1.show()
    S2.show()
```

Output :

Aryan

Rahul

⑤ what are attributes ?

→ Attributes are variable that store information about an object.

PYTHOD CODE:

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.mark = work
S1 = Student("Aryan", 85)
print(S1.name)
print(S2.mark
```

output:

Aryan
85

① what are methods)

→ methods are functions defined inside a class

PYTHON CODE:

```
class calculator:
    def add(self, a, b):
        return a+b
c= calculator()
print(c.add(5,3))
```

output: 8

⑦ Instance variables Vs class variables

Instance variable	class variable
→ Belongs to an individual object	→ Shared by the class
→ usually uses self	→ Defined directly inside the class
→ Each object can have a different value	→ One common value is used unless overridden

PYTHON code :

```
class Student :
    university = "MDX Dubai"
def __init__ (self, name):
    self.name = name
s1 = Student ("Aryan")
s2 = student ("Rahul")
print (s1.name)
print (s2.name)
print (s1.university)
```

Output:

```
Aryan
Rahul
MDX Dubai
```

⑧ List of objects.

→ we can store mutiple objects in a list

PYTHON CODE:

```
class student:
    def __init__(self, name):
        self.name = name
s1 = student("Aryan")
s2 = student("Rahul")
students = [s1, s2]
for s in students:
    print(s.name)
```

output:

Aryan
Rahul

⑨ Encapsulation:

Encapsulation means keeping data and the methods that work with it together inside a class.

PYTHON CODE:

```
class Student:
    def __init__(self, mark):
        self.__mark = mark
s1 = Student(85)
print(s1.__mark)
```

(10) Inheritance : Inheritance allows one class to reuse features of another class.

(11) Common Python Errors :

Error	Meaning
(1) eAttribute.Error	→ An attribute or method does not exist on the object.
(2) Type error	→ An operation or function an unsuitable type or arguments.
(3) Value error	→ A value has the right type but is unsuitable, such as int("hello")
(4) Name error	→ A variable or name has not been defined.